

Trevor Teolis

Postdoctoral Researcher, Rice University
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Research Interests

Partial differential equations, kinetic theory, interacting particle systems, collective dynamics, sampling methods, and machine learning.

Education

University of Illinois at Chicago, Chicago, IL

Ph.D., Mathematics, April 2025.

Dissertation: Well-posedness, long-time behavior, and multi-scale description of models of collective dynamics.

Advisor: Roman Shvydkoy.

University of Maryland, College Park, College Park, MD

B.S., Mathematics and Computer Science, May 2019.

Academic Positions

Rice University, Houston, TX

Postdoctoral Researcher, Aug 2025 – present.

Advisor: Maarten de Hoop.

University of Illinois at Chicago, Chicago, IL

Doctoral Researcher, Aug 2020 – May 2025.

Advisor: Roman Shvydkoy.

Publications

1. R. Shvydkoy, T. Teolis, “Microscopic, mesoscopic, and macroscopic descriptions of the Euler Alignment System with adaptive communication strength,” *Discrete and Continuous Dynamical Systems*, accepted (2025).
2. R. Shvydkoy, T. Teolis, “Well-posedness and long-time behavior of the Euler Alignment System with adaptive communication strength,” *Abel Symposium Proceedings*, accepted (2024).
3. P. Nandori, T. Teolis, “Local Equilibrium of Particle Density in Planar Lorentz Processes,” *Nonlinearity*, 2021.

Preprints

1. A. Balaji, T. Teolis, S. D. Mis, J. A. Lara Benitez, C. Wang, and M. V. de Hoop. “Hybrid operator learning of wave scattering maps in high-contrast media.” [arXiv:2602.11197](#), 2026.
2. R. Shvydkoy, T. Teolis, “Long-time behavior of the Fokker–Planck–Navier–Stokes Alignment System.” [arXiv:2508.07415](#), 2025.
3. A. Chertock, R. Shvydkoy, T. Teolis, “Modulation of the monokinetic limit for models of collective dynamics.” [arXiv:2508.05478](#), 2025.

Manuscripts in Preparation

1. T. Teolis and M. V. de Hoop. “Jump processes, stochastic in-context maps, mean-field limits, and the Boltzmann equation.”
2. T. Teolis, with N. Trillos, S. Li, J. Peszek, J. Maranzatto, S. Akkoc, K. Reidl, and S. Ulukus. “Rich Dynamics of Transformers.”
3. T. Teolis, S. D. Mis, and M. V. de Hoop. “FLOWERS for fast 3D Helmholtz forward modeling with applications to full-waveform inversion.” Manuscript in preparation.

Talks and Presentations

Invited Talks

- “Approximating Boltzmann solution maps by averaged measure-theoretic transformers,” minisymposium *Mathematical Foundations of Learning for Infinite-Dimensional Systems and PDEs*, SIAM Conference on Mathematics of Data Science, Salt Lake City, Utah, November 16–20, 2026. (upcoming)
- “Jump processes, stochastic in-context maps, mean-field limits, and the Boltzmann equation,” *Models of Emergence and Collective Dynamics*, 15th AIMS Conference, Athens, Greece, July 6–10, 2026. (upcoming)
- “Nambu particles as stochastic in-context maps for Boltzmann dynamics,” *Measure Flows for Inverse Problems and Machine Learning*, SIAM Conference on Uncertainty Quantification, Minneapolis, Minnesota, March 22–25, 2026.

Contributed Talks

- “The s-model and the unconditional alignment of particles in a fluid,” *Emergent Macroscopic Phenomena in Non-equilibrium Statistical Mechanics*, GSSI, L’Aquila, Italy, May 2025.
- “A new model of collective dynamics with adaptive communication strength,” *KinMAT Workshop*, University of Warsaw, June 2024.
- “Proof techniques for alignment and flocking,” IDEAL Seminar (online), March 2024.

Research Presentations to Funders

- “FLOWERS: A warp drive for neural PDE solvers,” research presentation to project funders, Oxy (Occidental Petroleum) Annual Project Review Meeting, Houston, Texas, May 19, 2026.
- “Interacting particle systems in machine learning and inference,” presentation to project reviewers from TGS, GMIG Annual Project Review Workshop, Rice University, Houston, Texas, September 16, 2025.

Poster Sessions

- “The s-model and the unconditional alignment of particles in a fluid,” *Workshop on Kinetic Theory and Fluids*, UW–Madison, March 2025.
- “The s-model and the unconditional alignment of particles in a fluid,” *Emergent Behavior in Complex Systems*, IMSI, Chicago, March 2025.

Departmental Seminars

- “Long-time behavior and multi-scale description of models of collective dynamics,” Ph.D. Defense, UIC, April 2025.
- “A new model of collective dynamics with adaptive strength,” Analysis and Applied Math Seminar, UIC, February 2024.

Awards

- Victor Twersky Memorial Award, University of Illinois at Chicago, 2024.
- Award for Graduate Research, University of Illinois at Chicago, 2024.
- Travel Award, University of Illinois at Chicago, 2024.

Workshops and Summer Schools

- Emergent Macroscopic Phenomena, GSSI, Italy, May 2025.
- Workshop on Kinetic Theory and Fluids, UW–Madison, March 2025.
- Emergent Behavior in Complex Systems, IMSI, Chicago, March 2025.
- Winter School: Boundary and Singularity in Fluid Mechanics, Stony Brook, Jan 2025.
- KinMAT Workshop, University of Warsaw, June 2024.
- Summer School: Fluid Dynamics, UMD, July 2023.
- Summer School: PDEs, Duke, Aug 2022.

Teaching

University of Illinois at Chicago

Teaching Assistant for Calculus I–III, Differential Equations, and Linear Algebra. Led discussion sections, mini-lectures, and office hours.

Professional Experience

Rampart Communications, Linthicum, MD

Summer Doctoral Intern, Jun – Aug 2024. Developed and implemented polar-coded modulation schemes.

Argonne National Laboratory, Lemont, IL

NSF Summer Intern, Jun – Aug 2021. Developed algorithms for mixed-integer PDE-constrained optimization.

National Institutes of Health, Bethesda, MD

Undergraduate Researcher and Software Engineer, Jan 2018 – Jul 2020. Built finite-element models of brain tumor growth using MRI and DTI data.

References available upon request.

Updated May 2026